



Saccharomyces cerevisiae improves rennet-induced gelation of ultra-high temperature milk

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ABSTRACT

Heating milk at high temperatures impairs its renneting properties, but rennet-induced curds can be formed from ultra-high temperature (UHT) milk inoculated with *Saccharomyces cerevisiae*. Herein, we measured physicochemical indices of UHT milk inoculated with *S. cerevisiae* before rennet addition, monitored the kinetics of gel formation, and investigated the physicochemical properties and microstructure of rennet-induced curds to explore the mechanisms by which *S. cerevisiae* influenced rennet-induced gelation of UHT milk. Compared with untreated pasteurized cow milk and UHT milk, the ethanol content was increased, the pH was decreased, the particle size and ζ -potential were increased, the time points at which the elasticity index began to increase were advanced, and the maximum elasticity index was increased for UHT milk inoculated with *S. cerevisiae*. The number of *S. cerevisiae* cells affected the structure of rennet-induced curds; with few cells added, the protein network of curds was continuous and tight, the mean square displacement curves showed an asymptotic behavior, and the water retention capacity and curd yield were high; with more cells added, the loosely entangled proteins aggregated, the continuity of the network was destroyed, and the curd yield decreased. In summary, a low number of *S. cerevisiae* cells ($<1.0 \times 10^7$ cfu/mL) can increase particle size, ζ -potential, and ethanol content, and decrease pH of *S. cerevisiae*-inoculated UHT milk, thereby accelerating the aggregation reactions after enzymatic reaction and improving the renneting properties.

Key words: *Saccharomyces cerevisiae*, ultra-high temperature milk, rennet-induced gelation, pH, ethanol content

INTRODUCTION

The rennet-induced gelation of milk is a crucial step in the production of cheese and has been widely studied for its important effects on product texture (Horne and Banks, 2004; Luo et al., 2017). During the primary stage of renneting, chymosin removes the N-terminal part of κ -casein from the surface of casein micelles, resulting in a significant increase in caseinomacropепptide (CMP) content and a decrease in electrostatic repulsion and steric stabilization of casein micelles. When enough CMP (85–90%) is cleaved, the second stage (aggregation of casein micelles) occurs, leading to formation of a gel network (Horne and Banks, 2004). The structure of rennet-induced gels depends on the physical and chemical properties of casein micelles (Zhao et al., 2014; Lin et al., 2017; Perreault et al., 2017).

Ultra-high temperature milk is produced by heating milk at high temperature (in the range of 135–150°C) for a short period of time (1–10 s; Deeth, 2020). Ultra-high temperature milk is widely used in acid-induced gelation (Nguyen et al., 2017; Gong et al., 2020) but not in rennet-induced gelation. It is well known that heating milk at temperatures exceeding 65°C results in changes to the physical and chemical properties of casein micelles, such as denaturation of whey proteins, which impairs the rennet-induced gelation of milk (Gamlath et al., 2018, 2020; Lin et al., 2018). Denatured whey proteins can form whey protein/ κ -casein complexes with κ -casein through disulfide bridges on the surface of casein micelles, or can form whey protein aggregates in serum (Kethireddipalli et al., 2011). The percentage of denaturation of whey proteins at high temperature depends on both temperature and time (Lin et al., 2018). Blecker et al. (2012) found that the renneting properties of milk diminished with increasing heat treatment, and the elastic modulus at the end of renneting was only 0.4 Pa when milk was heated for 20 min at 80°C.

In our previous study, considering the rich flavor characteristics in fermented milk inoculated with *Saccharomyces cerevisiae*, we attempted to produce cheese with novel flavors using these fermented milks (Li et

Received November 17, 2021.

Accepted April 8, 2022.

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Table 1. Preparation of *Saccharomyces cerevisiae*-inoculated UHT milk samples

Sample number	Raw milk	<i>S. cerevisiae</i> concentration (%, wt/vol)	Sucrose concentration (%, wt/vol)
UHT-3	UHT milk	0.02	3
UHT-4	UHT milk	0.02	4
UHT-5	UHT milk	0.02	5
UHT-6	UHT milk	0.02	6
UHT-7	UHT milk	0.02	7

al., 2021). Ultra-high temperature milk was inoculated with *S. cerevisiae* to prepare fermented milk, and the resulting milk was inoculated with starter cultures, coagulated with rennet, dehydrated, and pressed to obtain the cheese product. We found an interesting phenomenon, namely that rennet-induced curds could be formed in UHT milk inoculated with *S. cerevisiae*, and cheeses could be obtained from them. Thus, changes must occur in UHT milk inoculated with *S. cerevisiae*, leading to improved renneting properties of UHT milk, but the details remain unknown.

Therefore, the aim of the present work was to explore the mechanism by which *S. cerevisiae* influenced the rennet-induced gelation of UHT milk. Ultra-high temperature milk inoculated with *S. cerevisiae* was prepared, and the number of *S. cerevisiae* cells was modified by adjusting the amount of sucrose, with untreated pasteurized cow milk and UHT milk as controls. Before rennet-induced gelation, changes in the physicochemical properties of milk samples were characterized by measuring the pH, ethanol content, number of *S. cerevisiae* cells, particle size, and ζ -potential. The kinetics of renneting were probed using diffusing wave spectroscopy, and the microstructure of rennet-induced curds was observed using confocal laser scanning microscopy. Additionally, the composition and yield of the curds were determined. The results help explain why *S. cerevisiae* cells improve rennet-induced gelation of UHT milk, and provide theoretical support for the development of cheeses with novel flavors.

MATERIALS AND METHODS

Materials

Pasteurized cow milk containing 3.11% (wt/wt) protein and 3.74% (wt/wt) fat was obtained from Beijing Sanyuan Food Co. Ltd. and stored at 4°C until use. Ultra-high temperature milk containing 3.27% (wt/wt) protein and 3.83% (wt/wt) fat was obtained from Inner Mongolia Yili Industrial Group Co. Ltd. Sucrose was obtained from Guangzhou Fuzheng Donghai Food Co. Ltd. Active dry *S. cerevisiae* was obtained from Angel Yeast Co. Ltd. Rennet [Chy-Max Powder Extra NB,

2,235 international milk clotting units (IMCU)/g] was obtained from Chr. Hansen Inc. All other reagents were of analytical grade. No animals were used in this study, and ethical approval for the use of animals was thus deemed unnecessary.

Preparation of Milk Samples

According to Table 1, 5 UHT milk samples with different sucrose concentrations were mixed for 30 s at 20°C, inoculated with 0.02% (wt/vol) *S. cerevisiae* for 72 h at 23°C in a DSHZ-300A water bath incubator shaker (Suzhou Peiying Experimental Equipment Co. Ltd.). Untreated pasteurized cow milk and UHT milk served as controls and were named **P-0** and **UHT-0**, respectively. The milk samples were stored at 4°C for further use.

pH, Ethanol Content, and Yeast Count

The pH of milk samples was tested using a pH 211 microprocessor pH meter (Hanna Instruments), and the ethanol content and yeast count were measured according to the method of Kuda et al. (2016).

Particle Size and ζ -Potential

The particle size and ζ -potential of milk samples were measured using a Malvern Zetasizer Nano ZS ZEN3600 (Malvern Panalytical Ltd.) at 20°C according to the method of Zhang et al. (2019). Before measurement, samples were diluted with distilled water (1:800, vol/vol). Measurements were conducted at a scattering angle of 90°C, wavelength of 633 nm, and refractive index of water of 1.33.

Rennet-Induced Coagulation Measurement

For rennet-induced coagulation, milk samples were warmed for 20 min at 32°C, mixed with 0.02% (vol/vol) rennet, transferred to 20-mL glass cuvettes, and measured at 32°C using a Rheolaser Master diffusing wave spectroscopy instrument (Formulation Inc.). Data were collected every 1 min for 1 h. The elastic-

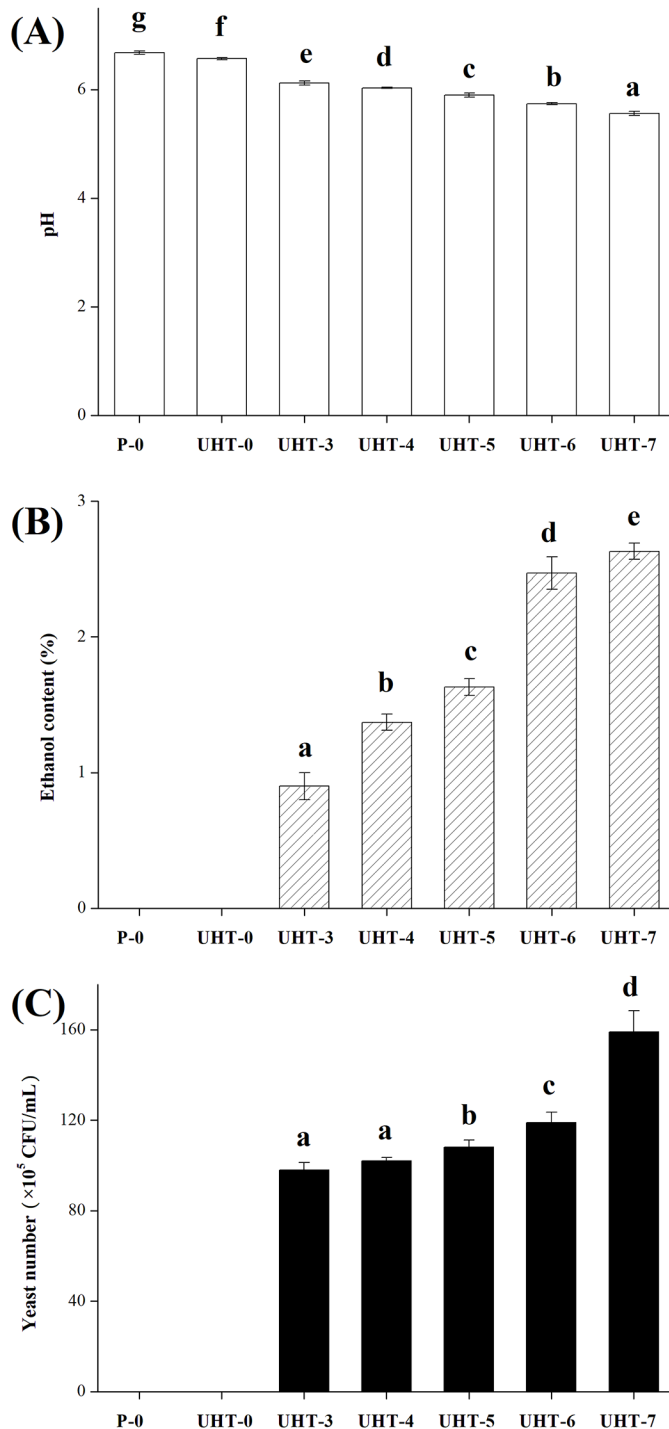


Figure 1. The pH (A), ethanol content (B), and number of yeast cells (C) in milk samples before rennet addition. P-0 = untreated pasteurized cow milk; UHT-0 = untreated UHT milk; UHT-3 = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 = *S. cerevisiae*-inoculated UHT milk with 7% sucrose. Each bar represents an average of each treatment ($n = 3$), and error bars represent SD for each mean. ^{a–g}Means with different letters are significantly different ($P < 0.05$).

ity index and mean square displacement (MSD) were analyzed using RheoSoft Master 1.4.0.0.

Curd Manufacture and Compositional Analysis

Milk samples were warmed for 20 min at 32°C and incubated with 0.02% (vol/vol) rennet for 1 h at 32°C. Resulting gels were cut into cubes of 1 cm, left to heal for 5 min, and centrifuged at $4,000 \times g$ for 25 min at 20°C. Curds were collected, and their yield, moisture content, and protein and fat levels were measured according to the method of Perreault et al. (2016).

Curd Microstructure

The microstructure of curds was observed via confocal laser scanning microscopy using an AiRsi instrument (Nikon Corp.) with a 100 $\times$ (oil) objective according to the method of Wang et al. (2019). Slices ($5 \times 5 \times 1$ mm) were taken from the middle of the curds, placed on a glass slide, and stained with 1 mL each of Fast Green (1 mg/mL, in water) and Nile Red (0.1 mg/mL, in ethanol) dyes. Dyes were absorbed for 5 min, excess dye was rinsed with distilled water, and a coverslip was applied. Subsequently, 2 laser excitation sources (633 and 488 nm) and 2 receiving channels were used to analyze Fast Green and Nile Red, respectively. Confocal laser scanning micrographs were analyzed using NIS-Elements Viewer 5.21.00 (Nikon Corp.).

Statistical Analysis

All experiments were repeated 3 times. Data were expressed as mean $\pm$ standard deviation and were analyzed by one-way ANOVA using SPSS 19.0 (IBM Corp.).

RESULTS

pH, Ethanol Content, and Yeast Count

Inoculation with *S. cerevisiae* had notable effects on the pH, ethanol content, and number of yeast cells in milk samples, as shown in Figure 1. Compared with P-0, the pH of UHT milk was decreased significantly ($P < 0.05$; Figure 1A). Compared with UHT-0, the pH of UHT milk inoculated with *S. cerevisiae* decreased significantly with increasing sucrose concentration ($P < 0.05$). Because *S. cerevisiae* and sucrose were not added to P-0 or UHT-0, the ethanol content and number of yeast cells were determined only in UHT milk inoculated with *S. cerevisiae* (Figures 1B and C). With increasing sucrose concentration, the ethanol content

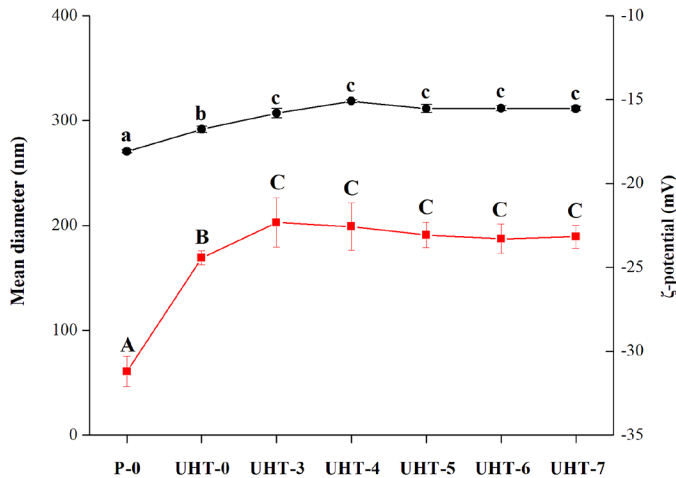


Figure 2. Mean diameter (●) and ζ -potential (■) in milk samples before rennet addition. P-0 = untreated pasteurized cow milk; UHT-0 = untreated UHT milk; UHT-3 = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 = *S. cerevisiae*-inoculated UHT milk with 7% sucrose. All results are expressed as mean $\pm$ SD ($n = 3$). ^{a-c}Means with different lowercase letters are significantly different in mean diameter ($P < 0.05$). ^{A-C}Means with different uppercase letters are significantly different in ζ -potential ($P < 0.05$).

and number of yeast cells in UHT milk inoculated with *S. cerevisiae* were increased significantly ($P < 0.05$).

Particle Size and ζ -Potential

Inoculation with *S. cerevisiae* had a notable effect on particle size and ζ -potential of milk samples, as shown in Figure 2. Compared with P-0, the particle size and ζ -potential of UHT milk were increased significantly ($P < 0.05$). Compared with UHT-0, the particle size and ζ -potential in UHT milk inoculated with *S. cerevisiae* were also increased significantly ($P < 0.05$). However, we found no significant differences between UHT milk samples inoculated with *S. cerevisiae* ($P > 0.05$).

Rennet-Induced Coagulation

Inoculation with *S. cerevisiae* had a notable effect on the elasticity index during renneting of milk samples, as shown in Figure 3 and Table 2. In P-0, the elasticity index remained unchanged at the beginning of renneting but increased steeply after 14 min, and then remained stable after 31 min. We observed no notable change in elasticity index in UHT-0 during the entire renneting period. In UHT milk inoculated with *S. cerevisiae*, the elasticity index was increased notably at the initial

stage of renneting, rapidly reaching stability and then remaining constant, except for UHT-7, which displayed a slight decrease. Compared with P-0, the time point at which the elasticity index began to increase and the earliest time point at which the elasticity index reached stability were significantly advanced in UHT milk inoculated with *S. cerevisiae* ($P < 0.05$), and the times were advanced more significantly with increasing sucrose concentration ($P < 0.05$). Compared with P-0, the maximum elasticity index and the elasticity index at the end of renneting were also increased significantly in UHT milk inoculated with *S. cerevisiae* ($P < 0.05$); these values increased first and then decreased with increasing sucrose concentration, and UHT-4 had the highest maximum elasticity index and the highest elasticity index at the end of renneting.

Inoculation with *S. cerevisiae* had a notable effect on the time-dependence behavior of MSD during renneting of milk samples, as shown in Figure 4. At the beginning of renneting (Figure 4A), the MSD curves for all groups showed obvious linear relationships, and no notable differences were detected between them. At the end of renneting (Figure 4B), the MSD curve was not significantly altered for UHT-0 ($P > 0.05$), and MSD curves retained a linear relationship for P-0, UHT-6, and UHT-7, but the curve slopes were significantly increased ($P < 0.05$). The MSD curves lost their linear relationship for UHT-3, UHT-4, and UHT-5, and instead displayed asymptotic behavior. Compared with P-0, the curve slopes at the end of renneting for UHT milk inoculated with *S. cerevisiae* first decreased and then increased with increasing sucrose concentration.

Curd Composition and Yield

Inoculation with *S. cerevisiae* had a notable effect on the composition and yield of rennet-induced curds, as shown in Table 3. No rennet-induced curd appeared in UHT-0. Compared with P-0, the curd yield and moisture content for UHT-3, UHT-4, and UHT-5 were increased significantly ($P < 0.05$), and the protein and fat levels were decreased significantly ($P < 0.05$). With a further increase in sucrose content, the yield and moisture content for UHT-6 and UHT-7 were decreased significantly ($P < 0.05$), and the protein and fat levels were increased significantly ($P < 0.05$).

Curd Microstructure

Inoculation with *S. cerevisiae* had a notable effect on the microstructure of rennet-induced curds, as shown in Figure 5. Rennet-induced curds in P-0 had a continuous and loose protein network, interrupted by numer-

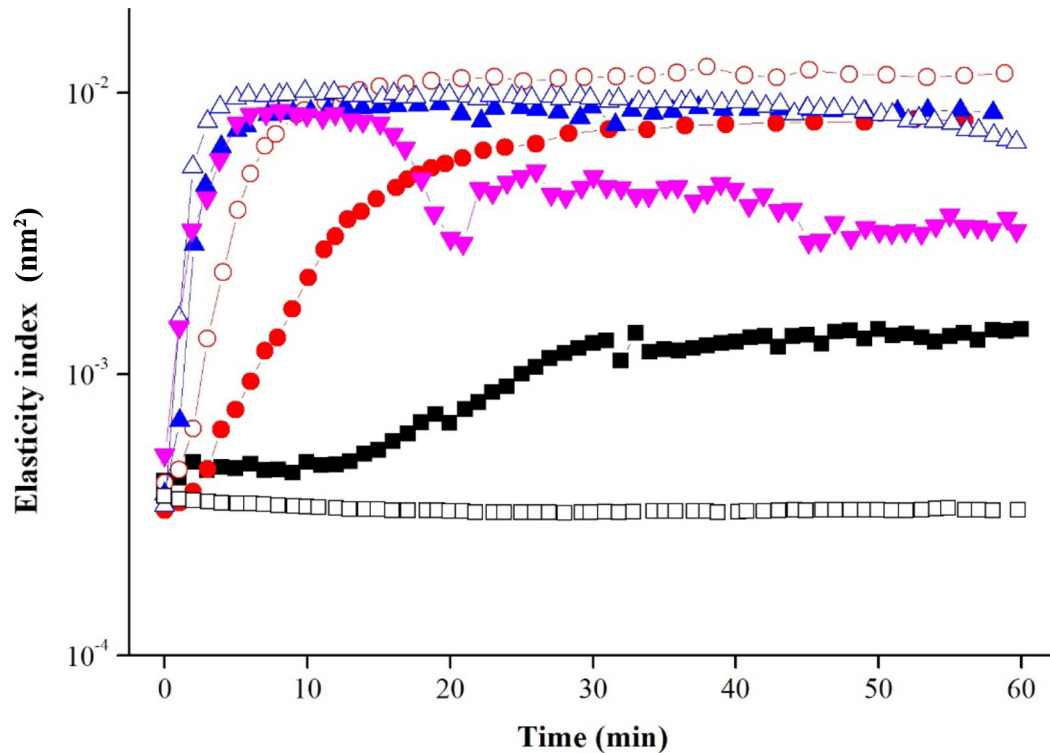


Figure 3. Elasticity index during renneting of milk samples. P-0 (■) = untreated pasteurized cow milk; UHT-0 (□) = untreated UHT milk; UHT-3 (●) = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 (○) = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 (▲) = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 (△) = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 (▼) = *S. cerevisiae*-inoculated UHT milk with 7% sucrose. Curves are representative runs.

ous uniform whey pockets and small, evenly dispersed fat globules. Compared with P-0, the protein networks in UHT-3, UHT-4, and UHT-5 remained continuous but were more compact; the numbers and sizes of whey pockets decreased; and fat globules were still evenly distributed in the protein networks, but these were enlarged, especially for UHT-5, which displayed notable

aggregation of fat globules. The integrity of the protein network was destroyed in rennet-induced curds in UHT-6 and UHT-7, with extensive protein aggregation. Additionally, numerous whey pockets of different sizes were dispersed throughout; fat globules were further enlarged, aggregated, and unevenly distributed; and the numbers of large fat globules were notably increased.

Table 2. Parameters during renneting of milk samples¹

Sample ²	T ₁ (min)	T ₂ (min)	EI _{max} (nm ²)	EI _{at 1 h} (nm ²)
P-0	13.79 ± 0.72 ^c	30.98 ± 0.99 ^d	0.0014 ± 0.0001 ^b	0.0013 ± 0.0001 ^b
UHT-0	—	—	0.0004 ± 0.0001 ^a	0.0004 ± 0.0001 ^a
UHT-3	2.59 ± 0.54 ^b	28.03 ± 1.22 ^c	0.0082 ± 0.0001 ^c	0.0079 ± 0.0002 ^e
UHT-4	1.26 ± 0.15 ^a	13.20 ± 0.61 ^b	0.0121 ± 0.0003 ^g	0.0116 ± 0.0002 ^g
UHT-5	<1	12.30 ± 0.78 ^b	0.0090 ± 0.0003 ^e	0.0086 ± 0.0001 ^f
UHT-6	<1	5.94 ± 0.98 ^a	0.0099 ± 0.0002 ^f	0.0068 ± 0.0002 ^d
UHT-7	<1	6.38 ± 0.62 ^a	0.0085 ± 0.0001 ^d	0.0033 ± 0.0002 ^e

^{a-g}Means with different letters within the same column are significantly different ($P < 0.05$).

¹All results are expressed as mean ± SD ($n = 3$). T₁ (min) = time point at which the elasticity index began to increase; T₂ (min) = earliest time point at which the elasticity index reached stability; EI_{max} = maximum elasticity index. —: not detected.

²P-0 = untreated pasteurized cow milk; UHT-0 = untreated UHT milk; UHT-3 = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 = *S. cerevisiae*-inoculated UHT milk with 7% sucrose.

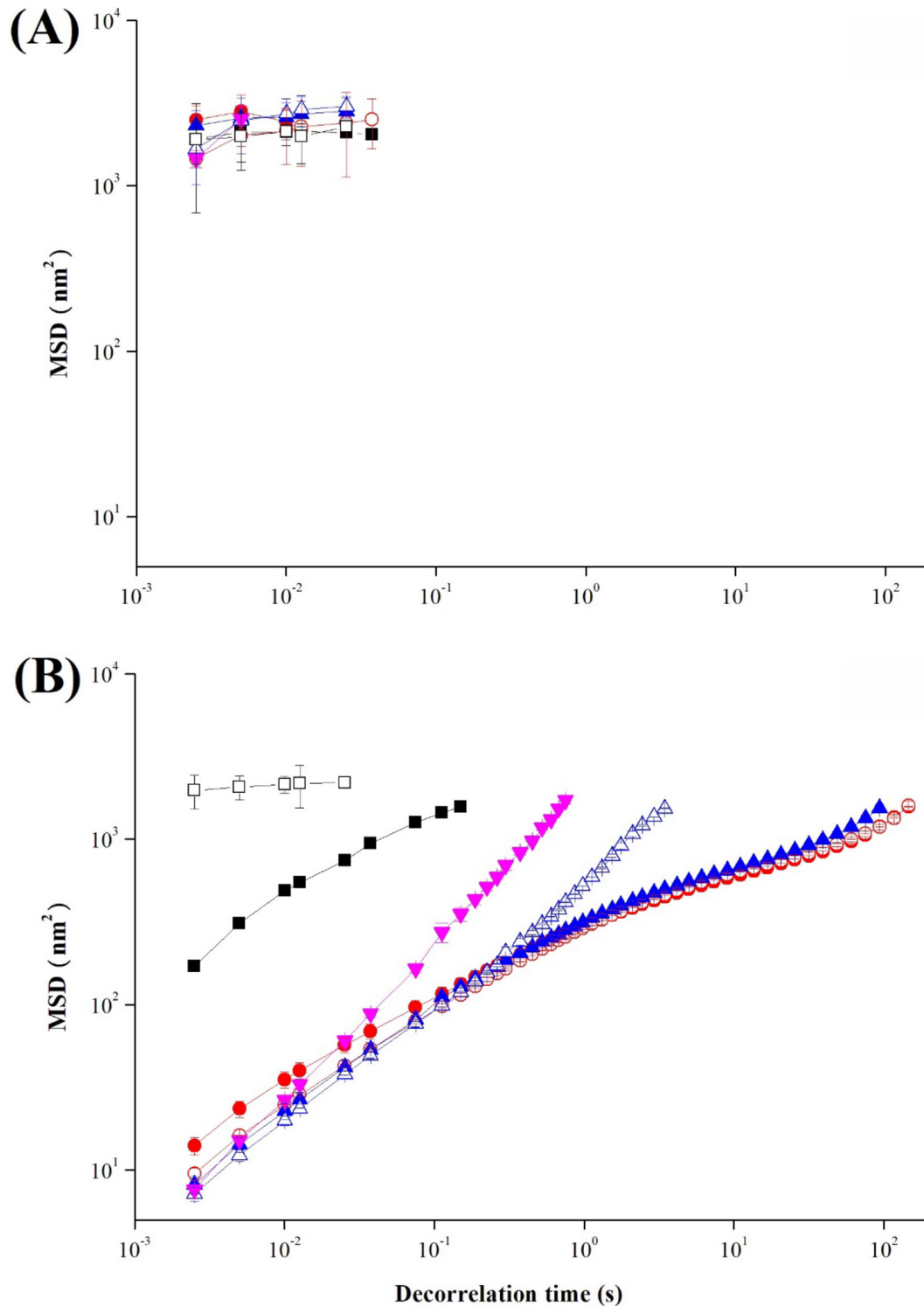


Figure 4. Time-dependence behavior of mean square displacement (MSD) during renneting of milk samples, (A) immediately after rennet addition and (B) at the end of renneting. P-0 (■) = untreated pasteurized cow milk; UHT-0 (□) = untreated UHT milk; UHT-3 (●) = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 (○) = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 (▲) = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 (△) = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 (▼) = *S. cerevisiae*-inoculated UHT milk with 7% sucrose. All results are expressed as mean $\pm$ SD (n = 3).

Table 3. Composition and yield of rennet-induced curds¹

Sample	Yield (%)	Moisture content (%)	Protein content (%)	Fat content (%)
P-0	27.62 ± 0.34 ^c	77.64 ± 0.81 ^c	9.00 ± 0.01 ^b	10.96 ± 0.33 ^c
UHT-0	—	—	—	—
UHT-3	33.25 ± 0.43 ^c	82.47 ± 0.67 ^d	7.73 ± 0.09 ^a	7.88 ± 0.52 ^a
UHT-4	32.76 ± 0.36 ^c	81.17 ± 0.93 ^d	7.98 ± 0.05 ^a	8.32 ± 0.47 ^a
UHT-5	28.38 ± 0.38 ^d	77.10 ± 1.53 ^c	9.11 ± 0.10 ^b	9.55 ± 0.50 ^b
UHT-6	23.32 ± 0.40 ^b	70.76 ± 0.46 ^b	11.20 ± 0.10 ^d	12.96 ± 0.53 ^d
UHT-7	21.42 ± 0.46 ^a	69.11 ± 0.48 ^a	10.46 ± 0.51 ^c	11.48 ± 0.37 ^c

^{a-c}Means with different letters within the same column are significantly different ($P < 0.05$).

¹All results are expressed as mean ± SD (n = 3). P-0 = untreated pasteurized cow milk; UHT-0 = untreated UHT milk; UHT-3 = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 = *S. cerevisiae*-inoculated UHT milk with 7% sucrose. —: not detected

DISCUSSION

P-0

The particle size and overall charge for P-0 were consistent with previous reports (de Kruif et al., 2012; Zhang et al., 2017). During rennet-induced gelation in P-0, rennet specifically acted on the Phe105-Met106 bond of κ -casein. When more than 85% CMP was re-

leased, casein micelles began to aggregate (Zhao and Corredig, 2020), causing an increase in elasticity index. At the initial stage of renneting (P-0), particles could move freely, and the MSD curve displayed a linear relationship, whereas at the end of renneting, the movement of particles was restricted by a uniform protein network, resulting in an increase in the curve slope. The moisture content and curd yield were higher in rennet-induced curds in P-0 than in conventional

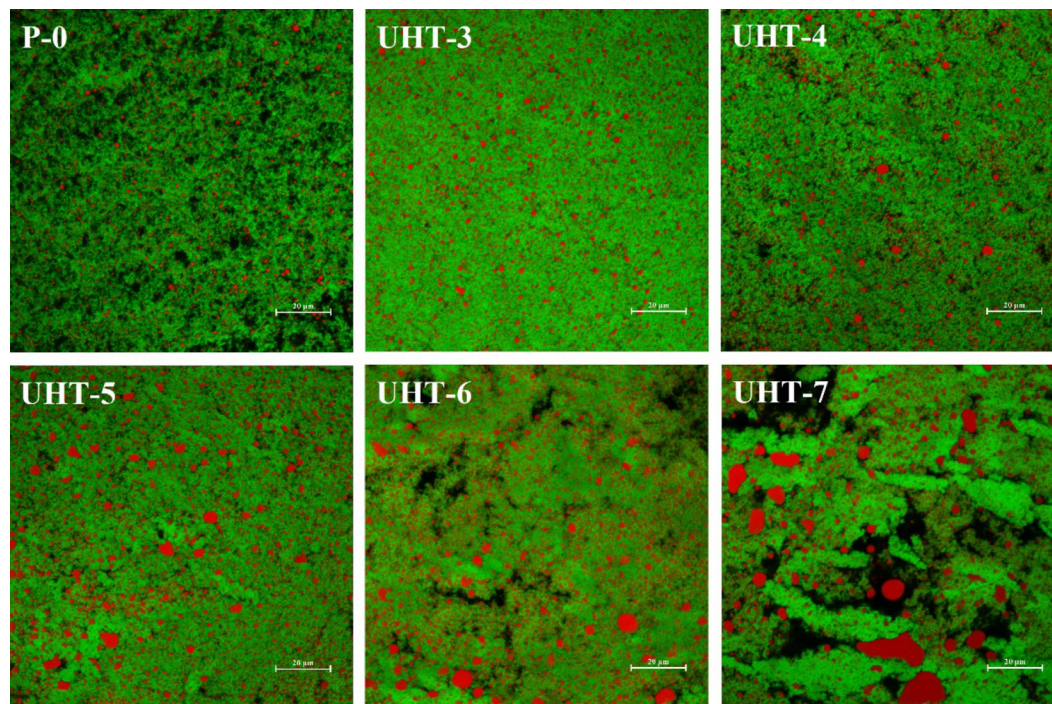


Figure 5. Confocal laser scanning micrographs of rennet-induced curds. Red = fat globules; green = proteins. P-0 = untreated pasteurized cow milk; UHT-3 = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 = *S. cerevisiae*-inoculated UHT milk with 7% sucrose. Images are representative runs.

cheese product, and the protein and fat contents were lower, which were related to low centrifugal force during curd manufacture.

UHT-0

Compared with P-0, the formation of whey protein/ κ -casein complexes or whey protein aggregates at high temperature led to an increase in particle size in UHT-0. Meanwhile, the binding effect would screen some negative charges, causing increased ζ -potential and decreased pH in UHT-0, as shown in the schematic diagram in Figure 6. Studies have shown that heat treatment has a more notable effect on the aggregation reaction of renneted casein micelles (paracasein micelles) than on the enzymatic reaction (Vasbinder et al., 2003; Blecker et al., 2012; Gamlath et al., 2018). The whey protein/ κ -casein complexes formed at high temperature can be completely hydrolyzed by rennet, but associated steric hindrance may impede the aggregation of renneted micelles (Blecker et al., 2012; Gamlath et al., 2018). Therefore, the elasticity index and MSD curves did not change significantly during the entire renneting period for UHT-0, and rennet-induced curd could not be formed.

UHT-3, UHT-4, and UHT-5

In UHT-3, UHT-4, and UHT-5, *S. cerevisiae* cells produced ethanol and carbon dioxide through anaerobic metabolism, resulting in increased ethanol content and decreased pH. With increasing sucrose, the number of *S. cerevisiae* cells increased, the ethanol content further increased, and pH value decreased further. Studies confirmed occurrence of no significant change in particle size in milk over a pH range of 5.8 to 6.7 (Lee and Lucey, 2010). Compared with P-0 and UHT-0, the increased particle size in UHT-3, UHT-4, and UHT-5 might be attributed to the increased ethanol content. The polarity of ethanol is lower than that of water, which would affect the structure of proteins by destroying non-covalent interactions (Feng et al., 2020; Peng et al., 2020). Thus, under the action of ethanol, the protein structure opened and became looser in UHT-3, UHT-4, and UHT-5, leading to an increase in particle size, as shown in Figure 6. Compared with P-0 and UHT-0, the decreased pH in UHT-3, UHT-4, and UHT-5 resulted in increased ζ -potential.

The renneting properties detected by diffusing wave spectroscopy were significantly improved in UHT-3, UHT-4, and UHT-5. As mentioned previously, rennet can act on the whey protein/ κ -casein complexes formed at high temperature, releasing CMP, as shown in Figure 6. Aggregation of paracasein micelles after enzymatic

action occurs via 2 steps: diffusion through the medium by Brownian motion, which enables particles to get closer to each other, and reactions between particles, which enables them to fuse together (Gamlath et al., 2018). According to previous studies, the decrease in milk pH would affect the structure of casein micelles, leading to progressive depletion of colloid calcium phosphate, which would accelerate the rate at which micelles aggregate (Li and Zhao, 2019; Liu et al., 2019; Zhao and Corredig, 2020). Additionally, compared with P-0 and UHT-0, the increased particle size, together with the decreased pH, also increased the diffusion and fusion rates of renneted casein micelles in UHT-3, UHT-4, and UHT-5, advancing the time point at which the elasticity index changed. The increased soluble calcium and decreased pH could enhance the elastic modulus during renneting (Li and Dalgleish, 2006; Zhao and Corredig, 2020), which might contribute to the higher elasticity index observed for UHT-3, UHT-4, and UHT-5 compared with P-0 and UHT-0. The MSD curves for UHT-3, UHT-4, and UHT-5 lost their linear relationship, indicating a stronger structure (Zhang et al., 2017), consistent with their higher elasticity index and tighter microstructure.

Under the combined actions of acid and rennet, rennet-dominated curds would be converted to acid-dominated curds with increasing starter amount, decreasing the syneresis of curds (Castillo et al., 2006). The pH values were lower in UHT-3, UHT-4, and UHT-5 than in P-0, and the combined action of acid and rennet might contribute to their higher moisture contents. Furthermore, denatured whey proteins are reported to have higher water retention capacity than native whey proteins (Perreault et al., 2016; Gamlath et al., 2020). Zhang et al. (2017) indicated that curds with smaller pores had higher water capillary force than native whey proteins and contained more water as a result. All of these properties would account for higher moisture content in UHT-3, UHT-4, and UHT-5 than in P-0. The increased moisture content led to high curd yield and low protein and fat contents in UHT-3, UHT-4, and UHT-5.

UHT-6 and UHT-7

Compared with other milk samples, the higher sucrose contents in UHT-6 and UHT-7 contributed to higher ethanol levels and lower pH. Previous studies have indicated that casein micelles are partly disintegrated into loosely entangled aggregates at pH 5.7 to 5.4 (Liu et al., 2019). With a further decrease in pH in UHT-6 and UHT-7, the depletion of colloid calcium phosphate and dissociation of proteins are increased (Liu et al., 2019; Zhao and Corredig, 2020), and whey protein/ κ -casein

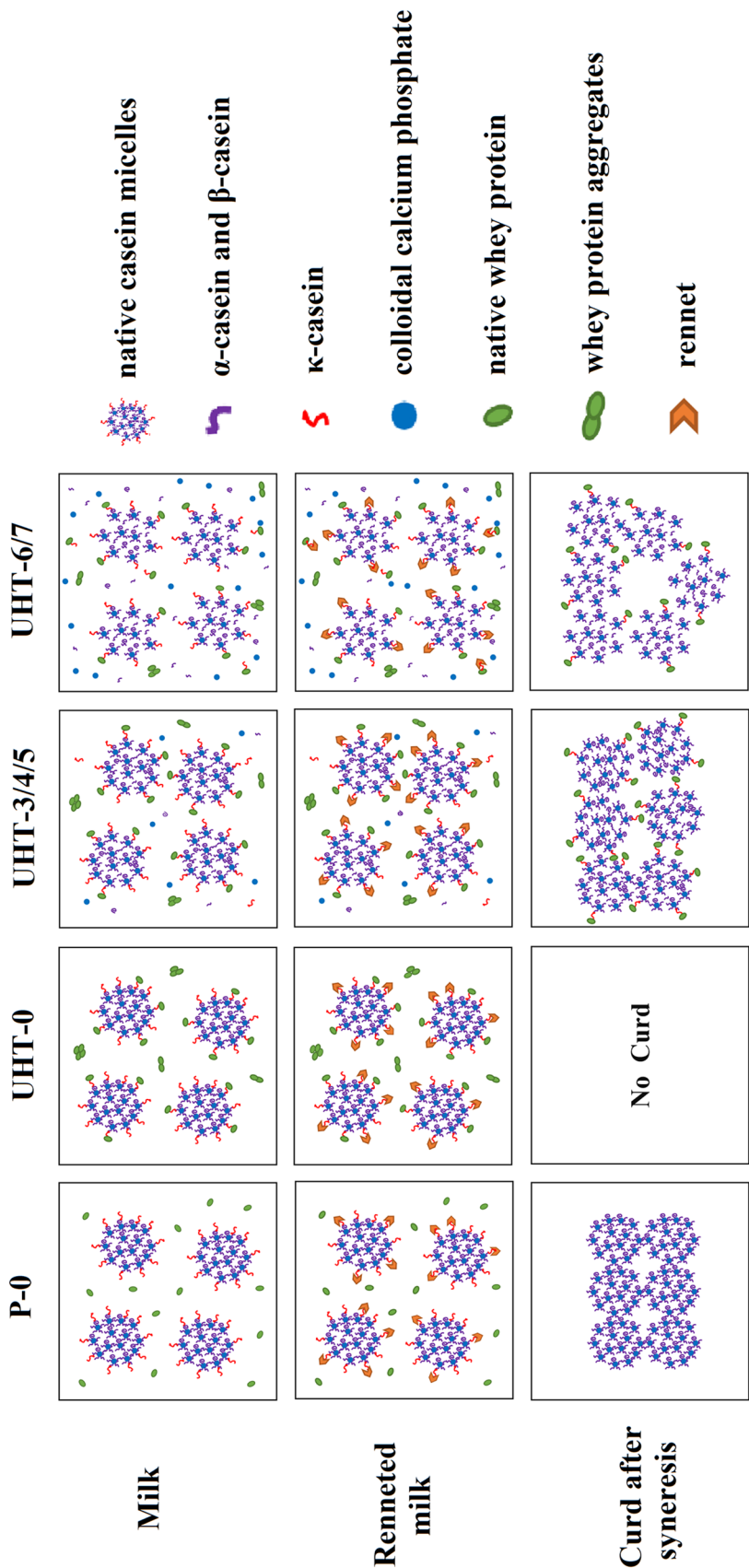


Figure 6. Schematic diagram outlining the proposed interpretation of experiments. P-0 = untreated pasteurized cow milk; UHT-0 = untreated UHT milk; UHT-3 = *Saccharomyces cerevisiae*-inoculated UHT milk with 3% sucrose; UHT-4 = *S. cerevisiae*-inoculated UHT milk with 4% sucrose; UHT-5 = *S. cerevisiae*-inoculated UHT milk with 5% sucrose; UHT-6 = *S. cerevisiae*-inoculated UHT milk with 6% sucrose; UHT-7 = *S. cerevisiae*-inoculated UHT milk with 7% sucrose.

complexes may also be dissolved (Liu et al., 2019). Thus, the particle sizes in UHT-6 and UHT-7 were not significantly different from those of UHT-3, UHT-4, and UHT-5, as shown in Figure 6. The unfolding of proteins generated more charged groups (Feng et al., 2020), and negative charges exposed would partly compensate for the neutralizing of charges caused by the decreased pH; hence we found no significant difference in ζ -potential among UHT milk samples inoculated with *S. cerevisiae*.

Compared with UHT-3, UHT-4, and UHT-5, the further decrease in pH and increase in soluble calcium in UHT-6 and UHT-7 made the paracasein micelles more likely to aggregate during renneting, advancing the time point at which the elasticity index changed. The elasticity index is related to the number and strength of bonds formed between proteins during renneting (Perreault et al., 2016; Zhang et al., 2017). Zhao and Corredig (2020) studied the effects of pH modification on rennet-induced gelation of concentrated casein micelle suspensions, and found that curds obtained at pH 6.0 had higher elastic modulus than those at pH 5.6, and suggested that the integrity of casein micelles played an important role in the network formation and its rheological properties. During renneting in UHT-6 and UHT-7, loose protein aggregates could form networks under the action of hydrophobic interactions, calcium bridges, and hydrogen bonds, but the formed networks were porous and weak, and the number and strength of bonds between proteins were decreased. Therefore, the elasticity index at the end of renneting was significantly lower in UHT-6 and UHT-7 than in UHT-3, UHT-4, and UHT-5, and the MSD curves returned to a linear relationship. The whey pockets were larger in UHT-6 and UHT-7 than in UHT-3, UHT-4, and UHT-5, indicating lower water retention capacity of curds, leading to decreased moisture content. A similar phenomenon was reported by Liu et al. (2014).

CONCLUSIONS

Compared with P-0, the renneting properties of UHT-0 were severely damaged; we observed no significant changes in elasticity index or MSD curves throughout the entire renneting period, and the rennet-induced curds could not be formed. *Saccharomyces cerevisiae* significantly improved the renneting properties of UHT milk, and the effects were dependent on the number of *S. cerevisiae* cells, which could be manipulated by adjusting the sucrose content. Compared with P-0 and UHT-0, in samples with relatively few *S. cerevisiae* cells (UHT-3, UHT-4, and UHT-5), the increased particle size and ethanol content and the decreased pH accelerated the aggregation of proteins, contributing to the formation of a tight, firm network with increased water

retention capacity. By contrast, in samples with more *S. cerevisiae* cells (UHT-6 and UHT-7), the further increase in ethanol content and decrease in pH caused the formation of loosely entangled protein aggregates, further accelerating the renneting process, but the networks formed were porous and weak, with decreased water capacity.

ACKNOWLEDGMENTS

This work was supported by the Program for Talents Introduction of Ludong University (Yantai, China) and the China Postdoctoral Science Foundation Project (Beijing; 2019M650282). We thank International Science Editing (<http://www.internationalscienceediting.com>) for editing this manuscript. The authors have not stated any conflicts of interest.

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